

Linear Models in R

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Introduction

The first half of the lab is a demonstration involving the Possums data. In the second half of the lab, you are asked to use what you learned in the demonstration to create and use a model for the `body_dims` dataset.

Getting set up

1. First, load the packages we will need:

```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(broom)
library(ggdists)
```

2. Next, we will load the possums dataset from our textbook's web page, <https://www.openintro.org/data/csv/possum.csv>

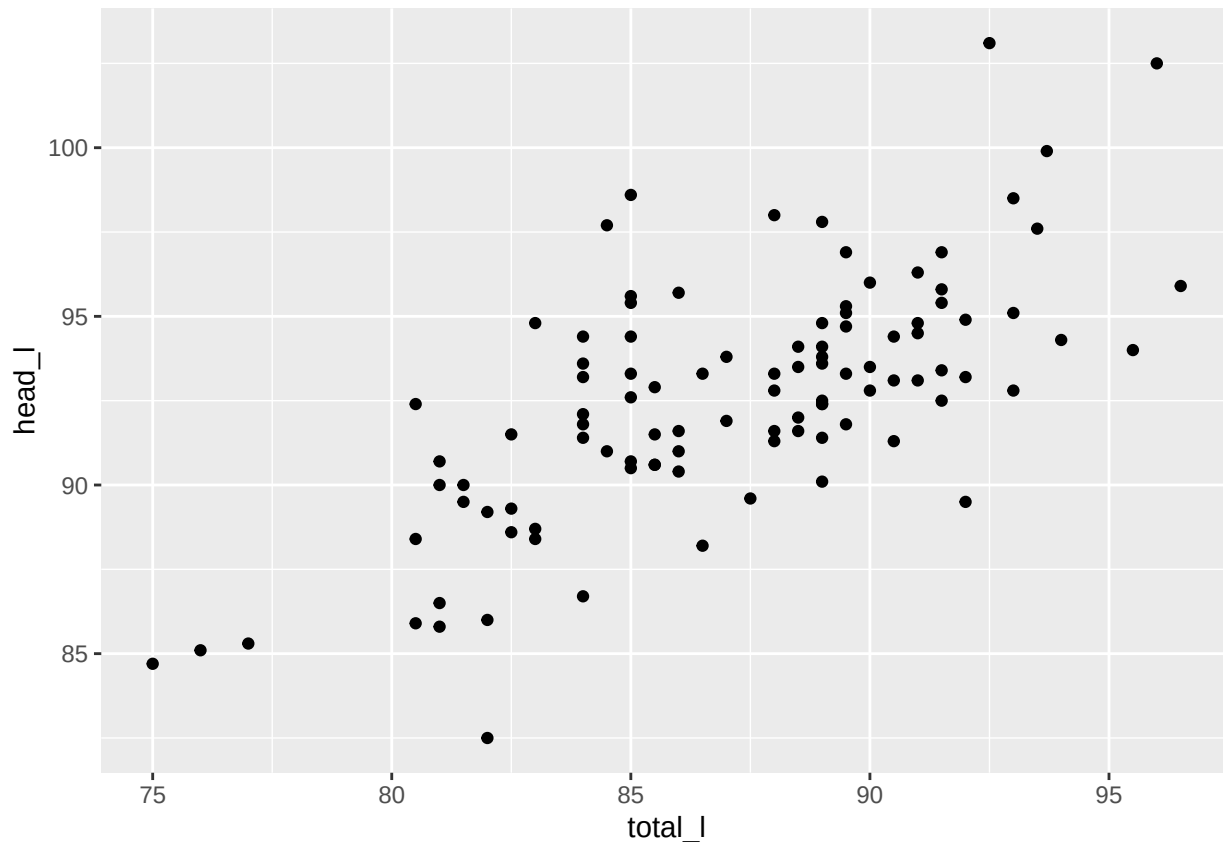
```
possum <- read_csv("https://www.openintro.org/data/csv/possum.csv")

## Rows: 104 Columns: 8
## -- Column specification -----
## Delimiter: ","
## chr (2): pop, sex
## dbl (6): site, age, head_l, skull_w, total_l, tail_l
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Graphing the data

3. We will reproduce the graph from the textbook here (but without circling one of points in red).

```
ggplot(data = possum,  
       mapping = aes(x = total_l,  
                     y = head_l)) +  
  geom_point()
```



Creating a model

Next, we learn the code to have R automatically compute a least squares regression model. You will use this code in one of our next activities. Least squares regression is a method for calculating a “line of best fit”. We will talk at a later time about how this method works.

We use the `lm` command (which stands for linear model) in R to calculate the least squares regression model. Look at the code chunk below.

ALERT: The order in which the variables are entered is important – the response variable is listed first and the explanatory (or predictor) variable is listed second. This may seem silly with just one explanatory variable, but scientists often make more complicated models where the response depends on many explanatory variables. So we can use this same structure to create a model to describe how the head length is modeled from not just total length, but also the mother’s head size, and the father’s head size, and whether the possum has some big head gene, etc like “`head_l~total_l+mother_head_l+father_head_l+big_head_gene`”

We are giving our new model the name `fit`.

4. Run the code chunk and notice that the model “fit” pops up in the Environment pane on the top right of your screen.

```
fit <- lm(head_l ~ total_l,
          data = possum)
```

5. We can ask R to display certain features of this model using the `summary()` command. Run the code chunk.

```
summary(fit)

##
## Call:
## lm(formula = head_l ~ total_l, data = possum)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.1877 -1.5340 -0.3345  1.2788  7.3968
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 42.70979     5.17281   8.257 5.66e-13 ***
## total_l      0.57290     0.05933   9.657 4.68e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.595 on 102 degrees of freedom
## Multiple R-squared:  0.4776, Adjusted R-squared:  0.4725
## F-statistic: 93.26 on 1 and 102 DF,  p-value: 4.681e-16
```

I can use this model summary output to write the EQUATION for the model. Here it is:

$$\text{head_l_hat} = 42.70979 + 0.57290 \cdot \text{total_l}$$

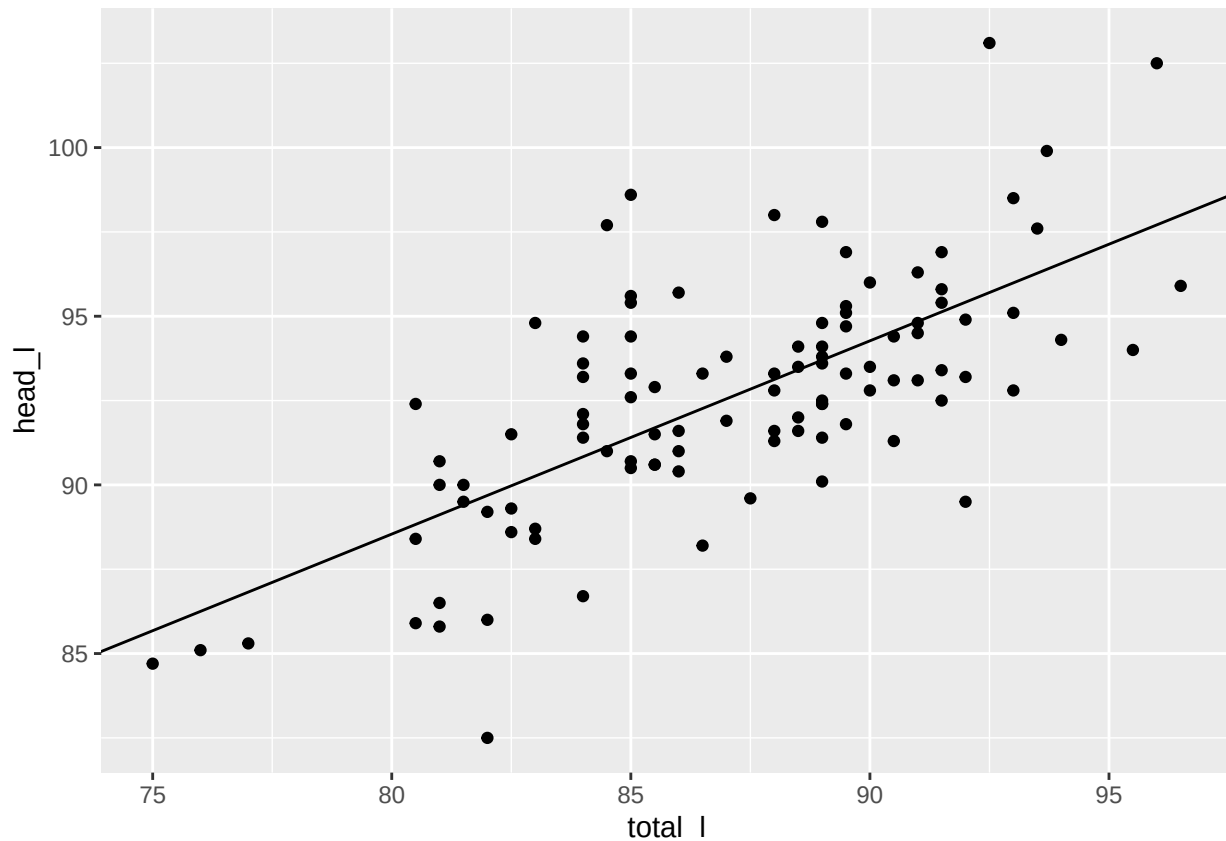
Here, 42.70979 is the intercept for the model and 0.57290 is the slope. These numbers are slightly different than the numbers our textbook authors used for their model.

6. Stop and examine the model summary output above, comparing it to the equation I wrote. Can you see where I found the numbers 42.70979 and 0.57290 in the output above? Make sure you find them as you will be writing another model equation on your own in a moment.

Graphing the model with the original data.

Next, I can graph the model together with the original data. The code is the same as before except that you will see I am adding a new layer `+geom_abline(intercept = 42.71, slope = 0.57)`.

```
ggplot(data = possum,
        mapping = aes(x = total_l,
                       y = head_l)) +
  geom_point() +
  geom_abline(intercept = 42.70979, slope = 0.57290)
```



Calculating predicted values and residuals

I can use R to calculate predicted values if I like. For example, to calculate the predicted head length for a possum that is 76cm in total length, I would use the model equation to calculate, in an R chunk:

```
42.70979 + 0.57290*76
```

```
## [1] 86.25019
```

There is a possum in the dataset that has total length 76cm and head length 85cm. I can calculate the residual for this datapoint using $\text{Residual} = \text{Actual} - \text{Predicted}$ as:

```
85 - 86.25019
```

```
## [1] -1.25019
```

Weight vs Height Part 2

8. Load the body_dims dataset.

Next, we load the body_dims dataframe we used in our last class and make the necessary conversions so that we have new variables called `height_in` and `weight_lb` recording the height in inches and weight in pounds, respectively, for the cases in the dataset.

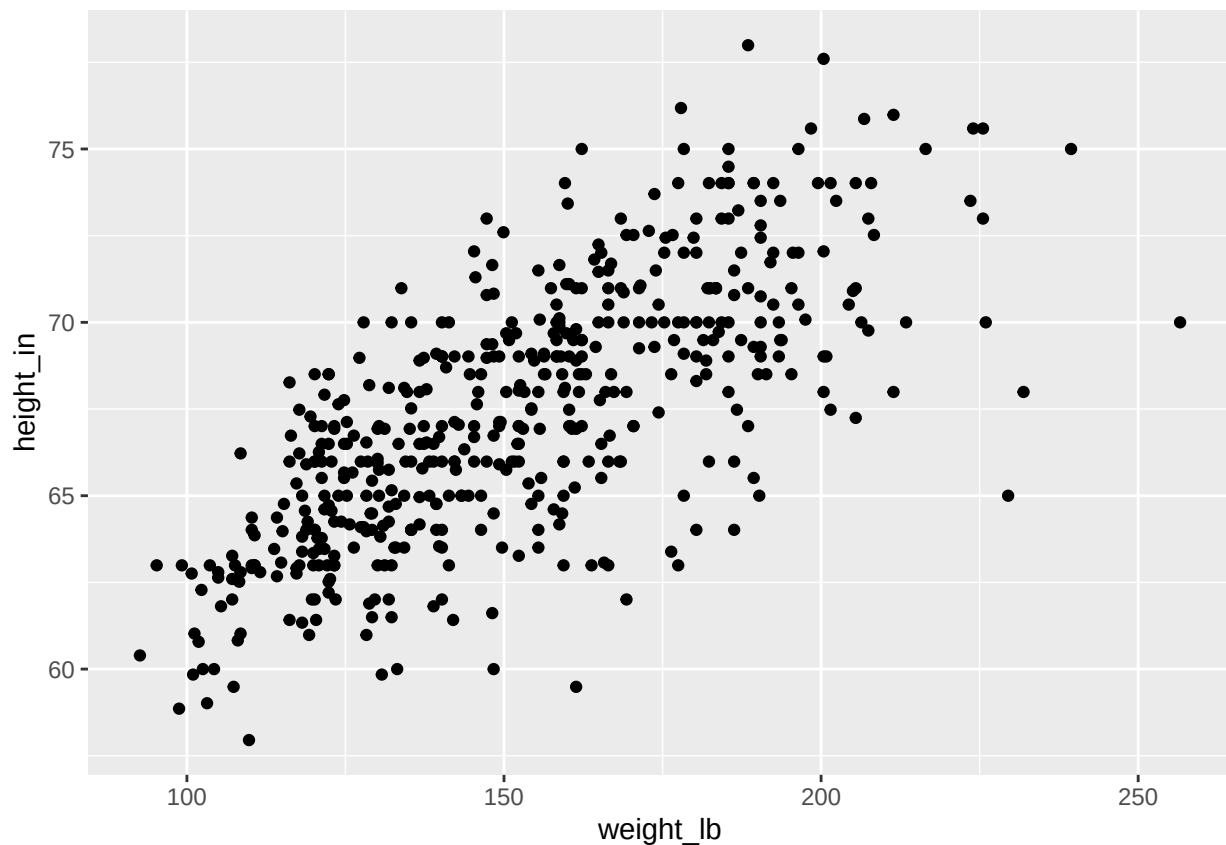
```
body_dims <- read_csv("https://www.openintro.org/data/csv/bdims.csv") %>%
  mutate(sex = factor(sex)) %>%
  mutate(height_in = hgt * 0.3937,
         weight_lb = wgt * 2.2046)
```

```
## Rows: 507 Columns: 25
## -- Column specification -----
## Delimiter: ","
## dbl (25): bia_di, bii_di, bit_di, che_de, che_di, elb_di, wri_di, kne_di, an...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Now, over to you.

9. Make a graph of weight in pounds versus height in inches.

```
ggplot(data = body_dims,
       mapping = aes(x = weight_lb,
                     y = height_in)) +
  geom_point()
```



10. Use `lm` to create a least squares regression model for predicting a person's weight from their height.

```
fit2 <- lm(weight_lb ~ height_in,
           data = body_dims)
```

11. Print the model summary output.

```
summary(fit2)
```

```
##
## Call:
## lm(formula = weight_lb ~ height_in, data = body_dims)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -41.322 -14.114  -2.714   11.152   90.615
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -231.5078     16.6214  -13.93  <2e-16 ***
## height_in     5.6983      0.2463   23.14  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.52 on 505 degrees of freedom
## Multiple R-squared:  0.5145, Adjusted R-squared:  0.5136
## F-statistic: 535.2 on 1 and 505 DF,  p-value: < 2.2e-16
```

12. Use the model summary output to write out the equation of the model line.

```
y_hat = -231.5078 + 5.6983 * height_in
```

13. (Optional.) You wrote interpretations of the slope and intercept for this model in the previous activity. If you would like to have them all in one place with this other information about this model, you can copy and paste those descriptions here.

Interpretation of Intercept: When the height (inches) is 0 inches, the predicted weight (pounds) is -231.5078 pounds.

Interpretation of the slope: For every 1 inch increase in height, the predicted weight increases by approximately 5.6983 pounds.

14. In the R chunk below, calculate the predicted weight for a person who is 5 feet tall.

```
predicted_weight <- -231.5078 + 5.6983 * 60
```

15. There is a person in the dataset who is 5 feet tall and weighs 133 pounds. In the R chunk below, calculate the residual for this case.

```
residual <- 133 - predicted_weight
```

16. What does the sign (positive or negative) of the residual tell you about this data point?

A positive residual indicates that the actual weight is higher than the predicted weight, meaning this data point is above the regression line. **### KNIT your file and upload for grading.**

If you are done before your classmates, help us out by copying and pasting any new code from this lab into our R cheatsheet. I have made a new slide at the end of the document.